

AFRILANG Stdlib

std/c/geomdistance

Français. Bibliothèque AFRILANG 'std/c/geomdistance' (niveau complexe, catégorie complexe). Elle expose 7 fonction(s) : euclidean2d, manhattan2d, chebyshev2d, pointToLine, pairwiseDist, centroid2d, bbox2d.

```
'import "std/c/geomdistance"' · 'use geomdistance'
```

- 'euclidean2d(x1 number, y1 number, x2 number, y2 number) → number' - 'manhattan2d(x1 number, y1 number, x2 number, y2 number) → number' - 'chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number' - 'pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) → number' - 'pairwiseDist(points list number) → list number' - 'c

English. AFRILANG library 'std/c/geomdistance' (tier complex, category complex). Exports 7 function(s): euclidean2d, manhattan2d, chebyshev2d, pointToLine, pairwiseDist, centroid2d, bbox2d.

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'import "std/c/geomdistance"' · 'use geomdistance'
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- 'euclidean2d(x1 number, y1 number, x2 number, y2 number) → number' - 'manhattan2d(x1 number, y1 number, x2 number, y2 number) → number' - 'chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number' - 'pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) → number' - 'pairwiseDist(points list number) → list number' - 'centroid2d(poi

Catégorie / Category	Modules complexe (c/)	
Niveau / Tier	complexe	June 16, 2026
Fonctions / Functions	7	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/geomdistance' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : euclidean2d, manhattan2d, chebyshev2d, pointToLine, pairwiseDist, centroid2d, bbox2d.

'import "std/c/geomdistance"' · 'use geomdistance'

```
- `euclidean2d(x1 number, y1 number, x2 number, y2 number) → number`
- `manhattan2d(x1 number, y1 number, x2 number, y2 number) → number`
- `chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number`
- `pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) →
  number`
- `pairwiseDist(points list number) → list number`
- `centroid2d(points list number) → list number`
- `bbox2d(points list number) → list number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/geomdistance"
use geomdistance
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- euclidean2d(x1 number, y1 number, x2 number, y2 number) → number
- manhattan2d(x1 number, y1 number, x2 number, y2 number) → number
- chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number
- pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) → number
- pairwiseDist(points list number) → list
- centroid2d(points list number) → list
- bbox2d(points list number) → list

4. Exemples d'utilisation

```
import "std/c/geomdistance"
use geomdistance
```

```
create result = euclidean2d(/* args */)
say result
```

```
create result = manhattan2d(/* args */)
say result
```

```
create result = chebyshev2d(/* args */)
say result
```

```
create result = pointToLine(/* args */)
say result
```

```
create result = pairwiseDist(/* args */)
say result
```

```
create result = centroid2d(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/geomdistance"`` en tête de fichier.
- Lancez ``afri-lang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module geomdistance

```
function euclidean2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function manhattan2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function chebyshev2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number
) returns number
end
```

```
function pairwiseDist(points list number) returns list number
end
```

```
function centroid2d(points list number) returns list number
end
```

```
function bbox2d(points list number) returns list number
end
end
```

English

1. Overview

AFRILANG library 'std/c/geomdistance' (tier complex, category complex). Exports 7 function(s): euclidean2d, manhattan2d, chebyshev2d, pointToLine, pairwiseDist, centroid2d, bbox2d.

'import "std/c/geomdistance"' · 'use geomdistance'

```
- `euclidean2d(x1 number, y1 number, x2 number, y2 number) → number`
- `manhattan2d(x1 number, y1 number, x2 number, y2 number) → number`
- `chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number`
- `pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) →
  number`
- `pairwiseDist(points list number) → list number`
- `centroid2d(points list number) → list number`
- `bbox2d(points list number) → list number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/geomdistance"
use geomdistance
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- euclidean2d(x1 number, y1 number, x2 number, y2 number) → number
- manhattan2d(x1 number, y1 number, x2 number, y2 number) → number
- chebyshev2d(x1 number, y1 number, x2 number, y2 number) → number
- pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number) → number
- pairwiseDist(points list number) → list
- centroid2d(points list number) → list
- bbox2d(points list number) → list

4. Usage examples

```
import "std/c/geomdistance"
use geomdistance
```

```
create result = euclidean2d(/* args */)
say result
```

```
create result = manhattan2d(/* args */)
say result
```

```
create result = chebyshev2d(/* args */)
say result
```

```
create result = pointToLine(/* args */)
say result
```

```
create result = pairwiseDist(/* args */)
say result
```

```
create result = centroid2d(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/geomdistance"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module geomdistance

```
function euclidean2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function manhattan2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function chebyshev2d(x1 number, y1 number, x2 number, y2 number) returns number
end
```

```
function pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number
) returns number
end
```

```
function pairwiseDist(points list number) returns list number
end
```

```
function centroid2d(points list number) returns list number
end
```

```
function bbox2d(points list number) returns list number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/geomdistance"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/geomdistance/>

Source (excerpt)

```
module geomdistance

function euclidean2d(x1 number, y1 number, x2 number, y2 number) returns number
end

function manhattan2d(x1 number, y1 number, x2 number, y2 number) returns number
end

function chebyshev2d(x1 number, y1 number, x2 number, y2 number) returns number
end

function pointToLine(px number, py number, x1 number, y1 number, x2 number, y2 number
    ) returns number
end

function pairwiseDist(points list number) returns list number
end

function centroid2d(points list number) returns list number
end

function bbox2d(points list number) returns list number
end
end
```